55. TUPLE-BASED TRANSACTION AUDIT

1. AIM

To store financial transaction records using tuples and audit them for duplicate transaction IDs, invalid amounts, and unapproved transaction types.

2. PROBLEM STATEMENT

Financial systems require automated audit mechanisms to ensure data integrity and detect anomalies such as duplicate transactions, zero or negative amounts, and invalid operation types. Write a Python program that accepts a list of transactions stored as tuples containing transaction ID, amount, and kind ('deposit' or 'withdrawal') and performs verification checks.

3. OBJECTIVES

- 1. Store transaction records as tuples inside a list data structure.
- 2. Iterate through records and unpack tuple values dynamically.
- 3. Track seen transaction IDs to detect and flag duplicate entries.
- 4. Validate transaction amounts (> 0) and transaction kinds ('deposit' or 'withdrawal').

4. ASSUMPTIONS & REQUIREMENTS

Assumptions:

- Transaction ID is treated as string input.
- Valid transaction types are strictly 'deposit' and 'withdrawal'.
- Amounts must be strictly positive floating-point values.

Requirements:

- Python 3.x Interpreter
- IDLE / PyCharm / VS Code environment

5. ALGORITHM

Step 1: Initialize empty list `transactions = []`.

Step 2: Read total number of transactions n.

Step 3: Loop `i` from 0 to `n-1`:

a. Read `tid`, `amount`, and `kind`.

b. Append tuple `(tid, amount, kind)` to `transactions`.

Step 4: Initialize empty list `seen = []`.

Step 5: For each tuple `t` in `transactions`:

a. Unpack `tid, amount, kind = t`.

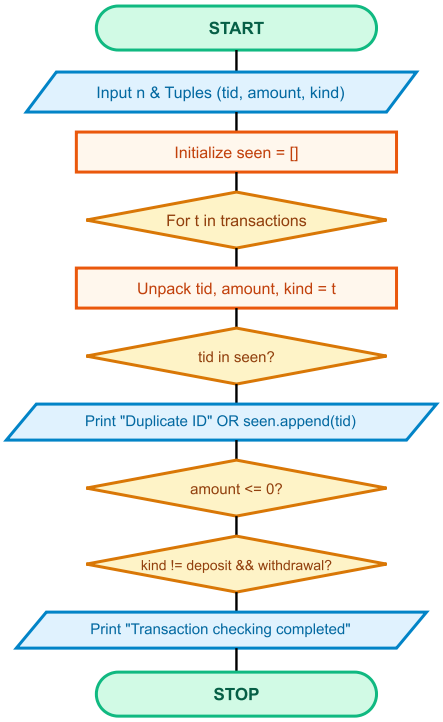
b. If `tid` in `seen`, print "Duplicate ID: <tid>", else append `tid` to `seen`.

c. If `amount <= 0`, print "Invalid amount: <tid>".

d. If `kind != "deposit"` and `kind != "withdrawal"`, print "Invalid transaction type: <tid>".

Step 6: Display "Transaction checking completed" and Stop.

6. PROGRAM FLOWCHART



7. PYTHON PROGRAM (TUPLE-BASED TRANSACTION AUDIT.PY)

```
transactions = [] n = int(input("Enter number of transactions: ")) for i in range(n)
```

8. EXPLANATION OF THE CODE

- **transactions.append((tid, amount, kind)):** Packs three input values into an immutable tuple and appends it to the list.
- **tid, amount, kind = t:** Unpacks tuple elements into individual variables for checking during iteration.
- **seen list:** Keeps track of previously encountered transaction IDs to detect duplicate entries.
- **if conditions:** Validates zero/negative transaction amounts and flags unauthorized transaction types.

9. TRACING TABLE

Transaction	Tuple (tid, amount, kind)	In Seen?	Amount > 0?	Kind Valid?	Audit Output
1	('1234', 500.0, 'deposit')	No (Appended)	Yes	Yes	None
2	('1234', 500.0, 'deposit')	Yes	Yes	Yes	Duplicate ID: 1234

10. TEST CASES

Test Case	Input Transactions	Expected Result	Actual Result
TC 1 (Duplicate ID)	n: 2 1) 1234, 500, deposit 2) 1234, 500, deposit	Duplicate ID: 1234 Transaction checking completed	Duplicate ID: 1234 Transaction checking completed
TC 2 (Invalid Amount/Type)	n: 1 1) 5678, -100, transfer	Invalid amount: 5678 Invalid transaction type: 5678 Transaction checking completed	Invalid amount: 5678 Invalid transaction type: 5678 Transaction checking completed

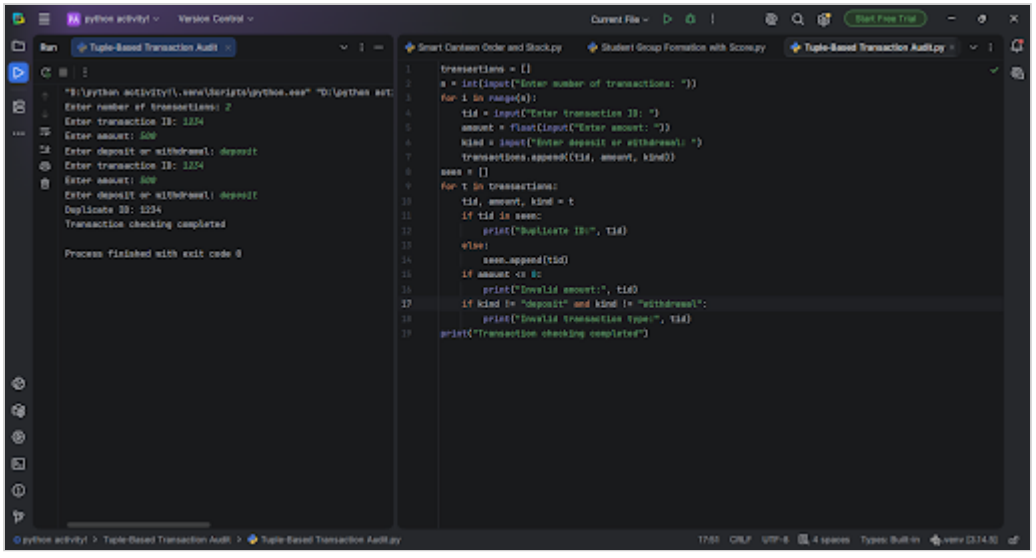
11. EXPECTED OUTPUT

```
Enter number of transactions: 2 Enter transaction ID: 1234 Enter amount: 500 Enter deposit
```

12. ACTUAL OUTPUT

```
Enter number of transactions: 2 Enter transaction ID: 1234 Enter amount: 500 Enter deposit
```

13. PROGRAM EXECUTION SCREENSHOT



14. RESULT/CONCLUSION

The Python program `Tuple-Based Transaction Audit.py` was executed successfully. It accurately audits financial records stored as tuples and flags duplicate transaction IDs, invalid amounts, and unapproved transaction kinds.